



Southeast University

Introduction to Software Engineering

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Outline

Part 0. Overview

Part 1. Software Processes

- ❖ **Chap 3. Software Process Structure**
- ❖ **Chap 4. Process Models**
- ❖ **Chap 5. Agile Development**
- ❖ **Chap 6. Human Aspects of Software Engineering**

Part 2. Modeling

Part 3. Quality Assurance

Part 4. Software Project Management

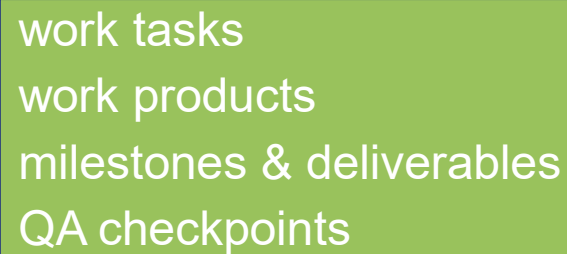
Chap 3. Software Process Structure

- ❖ **1. A Generic Process Model**
- ❖ **2. Identifying a Task Set**
- ❖ **3. Process Patterns**
 - **Process Pattern Types**
 - **CMMI (An example)**

Software Process Model

Process framework

Framework activities

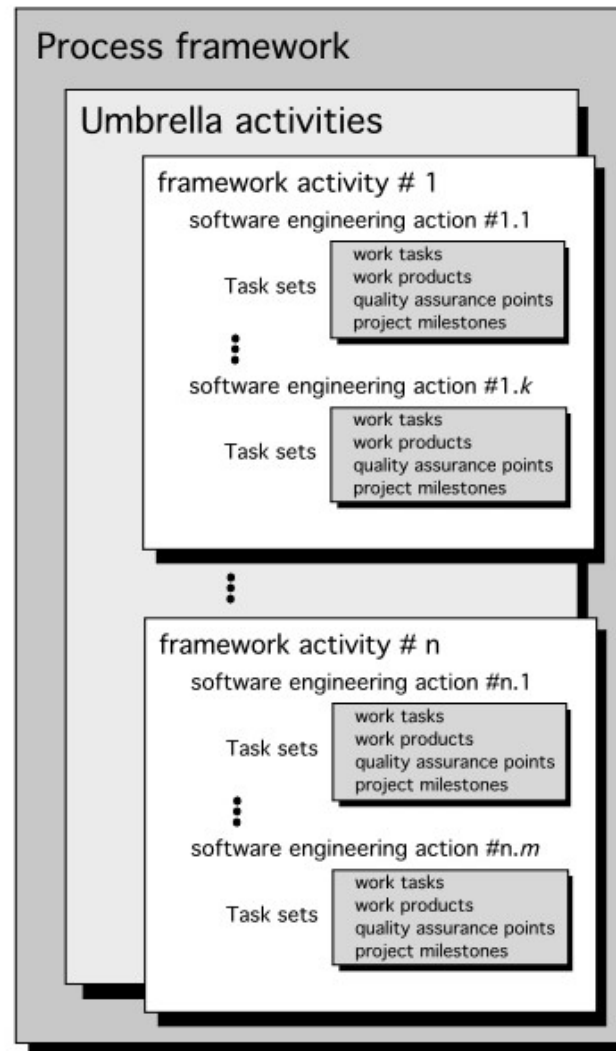


- work tasks
- work products
- milestones & deliverables
- QA checkpoints

Umbrella Activities

A Generic Process Model

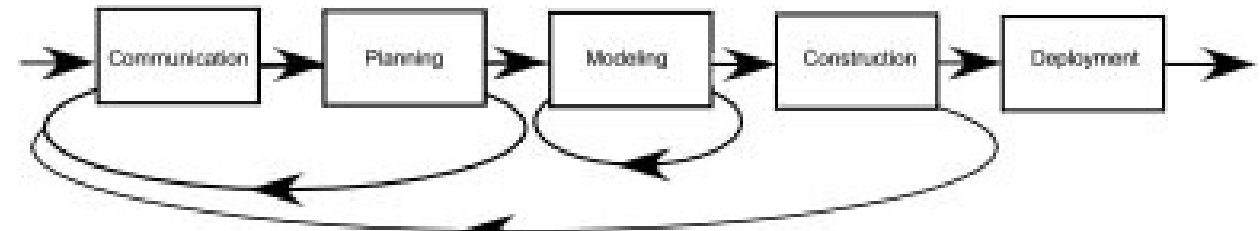
Software process



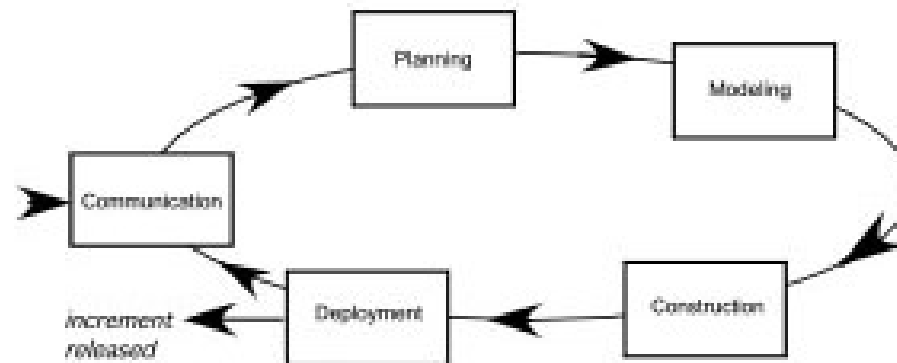
Process Flow



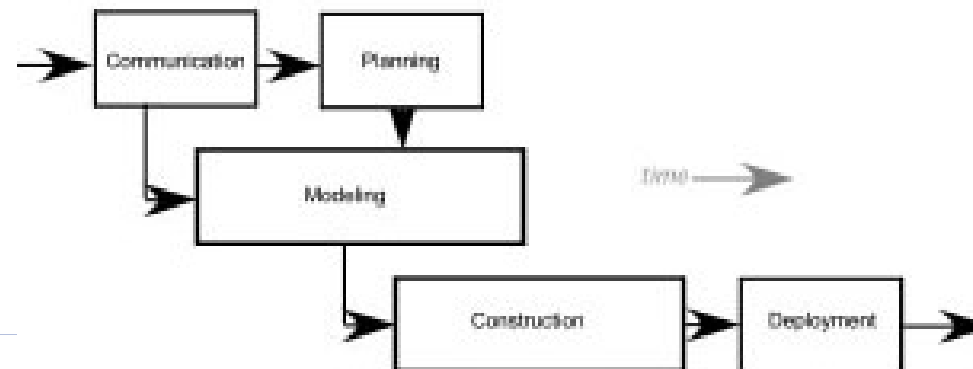
(a) linear process flow



(b) iterative process flow



(c) evolutionary process flow



(d) parallel process flow

Identifying a Task Set

- ❖ **A task set defines the actual work to be done to accomplish the objectives of a software engineering action.**
 - **A list of the task to be accomplished**
 - **A list of the work products to be produced**
 - **A list of the quality assurance filters to be applied**

- 明确任务集：定义了完成软件工程行动目标的实际工作

任务列表

需要生成的工作产品列表

需要应用的质量保证过滤器列表

CMMI

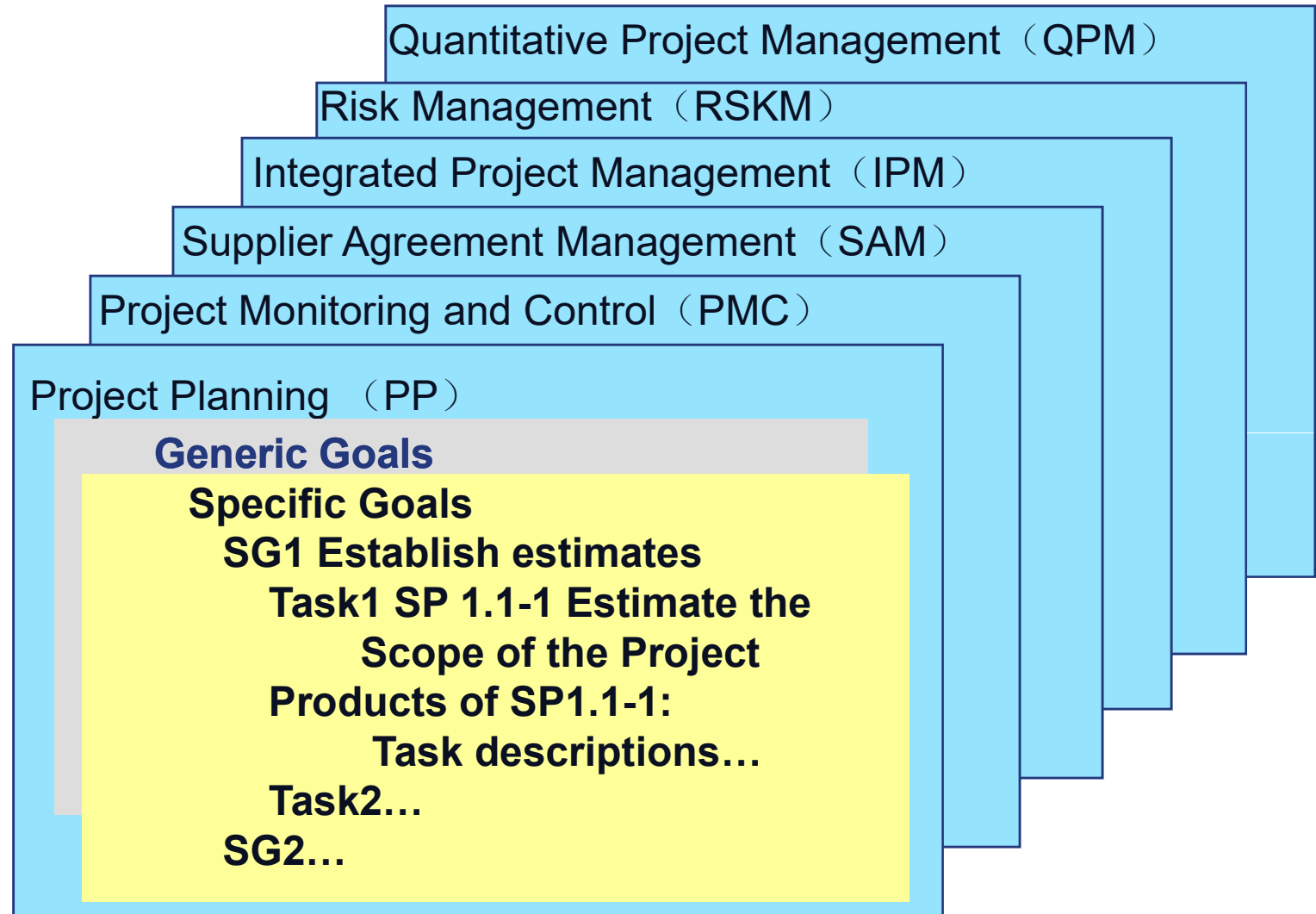
- ❖ **CMMI (Capability Maturity Model Integration) ——软件过程模型示例**
 - **A process meta-model proposed by the SEI (Software Engineering Institute).**
 - **CMMI is a collection of best practices that meet the needs of organizations in different areas of interest.**

CMMI ——软件过程模型示例
由SEI（软件工程研究所）提出的一个过程元模型。
CMMI 是满足不同领域内组织需求的最佳实践集合。

CMMI

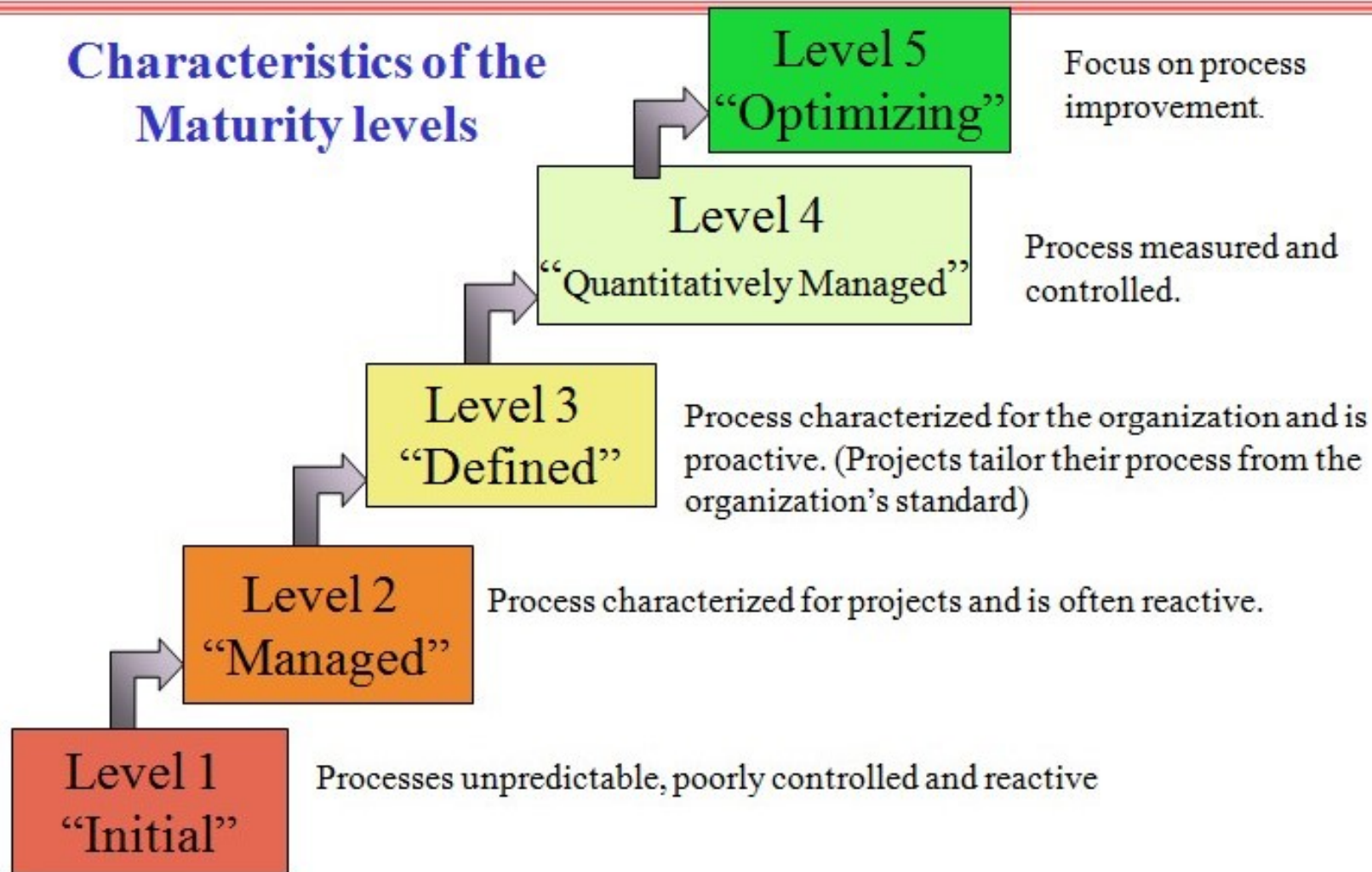
	Process Management	Project Management	Engineering	Support
1	—	—	—	—
2	—	PP,PMC, SAM	REQM	PPQA,CM, MA
3	OT,OPD, OPF	IPM, RSKM, QPM	RD, TS, PI,VER, VAL	DAR
4	OPP	—	—	—
5	OID	—	—	CAR

CMMI Example---Project Management



CMMI

Characteristics of the Maturity levels



• CMMI软件过程模型示例 ☆

1. **初始**：过程不可预测，缺乏管理
2. **已管理**：建立了基本的项目管理过程
3. **已定义**：软件过程标准化

4. **量化管理**：过程定量并控制
5. **持续优化**：关注过程改进

CMMI Stages

❖ 1.Initial(0)

❖ 2.Managed(7)

- **REQM**: Requirement Management
- **PP**: Project Planning
- **SAM**: Supplier Agreement Management
- **PMC**: Project Monitoring and Control
- **PPQA**: Process and Product Quality Assurance
- **CM**: Configuration Management
- **MA**: Measurement and Analysis

❖ 3.Defined(11)

- **OT , OPD , OPF , IPM , RSKM, RD , TS, PI , VER , VAL , DAR**

❖ 4. Quantitatively Managed(2)

- **QPM , OPP**

❖ 5.Optimized(2)

- **OID, CAR**

Software Process Model

❖ Process Assessment 软件过程评估

- The process should be assessed to ensure that it meets a set of basic process criteria that have been shown to be essential for a successful software engineering.
- Many different assessment options are available:
 - SCAMPI
 - CBA IPI
 - SPICE
 - ISO 9001:2000
 - Etc.

应对过程进行评估，以确保其满足一组基本的过程标准，这些标准已被证明对成功的软件工程至关重要



Process Assessment and Improvement

- ❖ **Standard CMMI Assessment Method for Process Improvement (SCAMPI)** — provides a five step process assessment model that incorporates five phases: initiating, diagnosing, establishing, acting and learning.
- ❖ **CMM-Based Appraisal for Internal Process Improvement (CBA IPI)**— provides a diagnostic technique for assessing the relative maturity of a software organization; uses the SEI CMM as the basis for the assessment [Dun01]
- ❖ **SPICE**—The SPICE (ISO/IEC15504) standard defines a set of requirements for software process assessment. The intent of the standard is to assist organizations in developing an objective evaluation of the efficacy of any defined software process. [ISO08]
- ❖ **ISO 9001:2000** for Software—a generic standard that applies to any organization that wants to improve the overall quality of the products, systems, or services that it provides. Therefore, the standard is directly applicable to software organizations and companies. [Ant06]

过程评估和改进

- 标准CMMI过程改进评估方法（SCAMPI）——提供了一种包含五个阶段的五步过程评估模型：启动、诊断、建立、执行和学习。
- 基于CMM的内部过程改进评估（CBA IPI）——提供了一种诊断技术，用于评估软件组织的相对成熟度；使用SEI CMM作为评估基础【Dun01】。
- SPICE——SPICE（ISO/IEC15504）标准定义了一套软件过程评估的要求。该标准的目的是帮助组织开发对任何已定义的软件过程的有效性的客观评估【ISO08】。
- ISO 9001:2000 for Software——一个通用标准，适用于任何希望提高其提供的产品、系统或服务的整体质量的组织。因此，该标准直接适用于软件组织和公司【Ant06】。

适应过程模型

Adapting a Process Model

❖ Adaptable Model

- The framework activities will always be applied on every project ... BUT
框架活动将始终应用于每个项目
- The tasks (and degree of rigor) for each activity will vary based on:
 - the type of project
 - characteristics of the project
 - common sense judgment; concurrence of the project team

每项活动的任务（和严格程度）将根据以下因素有所不同：
项目类型； 项目的特征；
常识判断；项目团队的一致意见

Adapting a Process Model

适应过程模型的因素

- the overall flow of activities, actions, and tasks and the interdependencies among them
- the degree to which actions and tasks are defined within each framework activity
- the degree to which work products are identified and required
- the manner which quality assurance activities are applied
- the manner in which project tracking and control activities are applied
- the overall degree of detail and rigor with which the process is described
- the degree to which the customer and other stakeholders are involved with the project
- the level of autonomy given to the software team
- the degree to which team organization and roles are prescribed

个人软件过程

Personal Software Process (PSP)

计划 ❖ **Planning.** This activity isolates requirements and develops both size and resource estimates. In addition, a defect estimate (the number of defects projected for the work) is made. All metrics are recorded on worksheets or templates. Finally, development tasks are identified and a project schedule is created.

高层设计 ❖ **High-level design.** An external specification is created for each component and a component design is created. Prototypes are built when uncertainty exists. All issues are recorded and tracked.

高层设计评审 ❖ **High-level design review.** Formal verification methods (Chapter 21) are applied to uncover errors in the design. Metrics are maintained for all important tasks and work results.

开发 ❖ **Development.** The component level design is refined and reviewed. Code is generated, reviewed, compiled, and tested. Metrics are maintained for all important tasks and work results.

事后分析 ❖ **Postmortem.** Using measures and metrics collected the effectiveness of the process is determined. If this is a large amount of data it should be analyzed statistically), Measures and metrics should provide guidance for modifying the process to improve its effectiveness.

团队软件过程

Team Software Process (TSP)

- ❖ **Build self-directed teams that plan and track their work, establish goals, and own their processes and plans. These can be pure software teams or integrated product teams (IPT) of three to about 20 engineers.**
- ❖ **Show managers how to coach and motivate their teams and how to help them sustain peak performance.**
- ❖ **Accelerate software process improvement by making CMM Level 5 behavior normal and expected.**
 - **The Capability Maturity Model (CMM), a measure of the effectiveness of a software process, is discussed in Chapter 30.**
- ❖ **Provide improvement guidance to high-maturity organizations.**
- ❖ **Facilitate university teaching of industrial-grade team skills.**



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Thank You !

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